

An exceptional four-dimensional unitary Hecke Yang–Baxter operator

A minimal ordinary localization for the $\mathcal{C}(\mathfrak{sl}_3, 6)$ Jones–Wenzl representations

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27 July 2026

Abstract

We give an exact 16×16 unitary Yang–Baxter operator in the exceptional class

$$\left[e^{i\pi/3}, \frac{1}{2}, 4 \right].$$

This class was the first unresolved case in Lechner’s 2026 classification of unitary two-eigenvalue R -matrices and is the ordinary-localization problem studied by Galindo, Hong, and Rowell for the $\mathcal{C}(\mathfrak{sl}_3, 6)$ Jones–Wenzl representations. Writing a four-dimensional site as two qubits, the solution is a sum of five real Pauli words. Four words combine into a reflection M , a fifth reflection E anticommutes with M , and the Yang–Baxter equation selects

$$H = \sqrt{\frac{2}{3}} M - \frac{1}{\sqrt{3}} E$$

from the circle of involutions $\alpha M + \beta E$. The spectral projection has both partial traces equal to $2\mathbf{1}_4$. Consequently the induced tensor-space representation has the $\eta = 1/2$ Markov trace and is faithful on every quotient $H_n(3, 6)$. This supplies an ordinary unitary localization of local dimension four; dimensions below four are impossible. A coordinate permutation also identifies the active 8×8 part as a faithful $(3, 2)$ -generalized localization. Two independent exact verifiers accompany the construction, one using direct sparse matrices over $\mathbb{Q}(\sqrt{2}, \sqrt{3}, i)$ and one using only the finite Pauli-word algebra.

Status. This is an unrefereed, AI-assisted research note. The proof, exact certificates, and source audit are public for independent checking; they have not received expert peer review. The priority conclusion is necessarily provisional.

1 Introduction and main result

Let V be a finite-dimensional complex Hilbert space. A unitary $R \in \text{End}(V \otimes V)$ is a Yang–Baxter operator when

$$(R \otimes \mathbf{1}_V)(\mathbf{1}_V \otimes R)(R \otimes \mathbf{1}_V) = (\mathbf{1}_V \otimes R)(R \otimes \mathbf{1}_V)(\mathbf{1}_V \otimes R). \quad (1)$$

Lechner [3] classified the possible characters of non-involutive unitary R -matrices with two eigenvalues, up to one exceptional family. In his normalization the two eigenvalues are -1 and q , the (-1) -spectral projection P has normalized trace

$$\eta = \frac{\text{rank } P}{(\dim V)^2},$$

and the unresolved classes are

$$\left[e^{i\pi/3}, \frac{1}{2}, 2d \right].$$

The class in base dimension two is empty; base dimension four is the first open case.

This question traces back to the case study of Galindo, Hong, and Rowell [2, §5.6]. They showed that the Jones–Wenzl representations associated with $\mathcal{C}(\mathfrak{sl}_3, 6)$ have a unitary quasi-localization and an 8×8 $(3, 1)$ -generalized localization, but no two-dimensional ordinary unitary localization. They wrote that the sequence did not appear to have an ordinary localization and noted it as a possible counterexample to the localization conjecture. Lechner’s work later isolated the precise matrix family above.

The following is the central result of this note.

Theorem 1.1. *Let*

$$q = e^{i\pi/3} = \frac{1 + i\sqrt{3}}{2}.$$

There is an explicit unitary $R \in \text{End}(\mathbb{C}^4 \otimes \mathbb{C}^4)$ such that

$$(R + \mathbf{1}_{16})(R - q\mathbf{1}_{16}) = 0, \quad R_1 R_2 R_1 = R_2 R_1 R_2,$$

and each eigenvalue $-1, q$ has multiplicity eight. Its (-1) -spectral projection P satisfies

$$\text{Tr}_1(P) = \text{Tr}_2(P) = \mathbf{21}_4.$$

Therefore

$$\left[e^{i\pi/3}, \frac{1}{2}, 4 \right] \neq \emptyset,$$

and the associated tensor-space representations induce faithful embeddings

$$H_n(3, 6) \hookrightarrow \text{End}((\mathbb{C}^4)^{\otimes n}) \quad (n \geq 2).$$

Under the identification in [2, §5.6], these embeddings form an ordinary unitary localization of the $\mathcal{C}(\mathfrak{sl}_3, 6)$ Jones–Wenzl representation sequence.

Corollary 1.2. *The minimum local dimension of an ordinary unitary localization of this Jones–Wenzl sequence is four.*

The formula for R is given in Section 2; the matrix identities are proved in Sections 3 and 4. Section 5 proves the full localization claim rather than only the two-site spectral statement. The minimality argument is in Section 6. Section 7 records a generalized Yang–Baxter normal form, and Section 9 separates the exact theorem from the necessarily incomplete priority audit.

2 The five-word construction

Put $V = \mathbb{C}^2 \otimes \mathbb{C}^2 \cong \mathbb{C}^4$, and define the real matrices

$$I = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}, \quad X = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad Z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}, \quad J = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}.$$

A word such as $ZIZZ$ denotes $Z \otimes I \otimes Z \otimes Z$ on $V \otimes V$. Define

$$H = -\frac{1}{\sqrt{6}}ZIZZ - \frac{1}{\sqrt{6}}ZIJJ - \frac{1}{\sqrt{6}}JIZJ + \frac{1}{\sqrt{6}}JIJZ - \frac{1}{\sqrt{3}}XIXX, \quad (2)$$

and

$$R = \frac{q-1}{2}\mathbf{1}_{16} + \frac{q+1}{2}H = \frac{-1+i\sqrt{3}}{4}\mathbf{1}_{16} + \frac{3+i\sqrt{3}}{4}H. \quad (3)$$

Equations (2)–(3) are an exact specification of all 256 matrix entries in the computational basis.

The relevant multiplication table is

$$\begin{aligned} X^2 &= Z^2 = I, & J^2 &= -I, \\ XZ &= J = -ZX, & XJ &= Z = -JX, & JZ &= X = -ZJ. \end{aligned}$$

Thus every product of tensor words reduces to a signed word over $\{I, X, Z, J\}$.

3 A circle of reflections

The five terms in (2) have a compact algebraic explanation. Set

$$A = ZIZZ, \quad B = ZIJJ, \quad C = JIZJ, \quad D = JIJZ, \quad E = XIXX$$

and

$$M = \frac{-A - B - C + D}{2}. \quad (4)$$

Lemma 3.1. *The operators A, B, C, D, E are real symmetric involutions. The first four commute, $D = ABC$, and E anticommutes with each of A, B, C, D . Consequently*

$$M^2 = E^2 = \mathbf{1}_{16}, \quad ME = -EM.$$

For real α, β ,

$$H(\alpha, \beta) = \alpha M + \beta E$$

satisfies

$$H(\alpha, \beta)^2 = (\alpha^2 + \beta^2)\mathbf{1}_{16}.$$

Proof. The word multiplication table gives every stated commutation relation and $D = ABC$. One may also prove $M^2 = \mathbf{1}$ without expanding. Simultaneously diagonalize A, B, C , and let their joint eigenvalues be $x, y, z \in \{\pm 1\}$. Since $D = ABC$, the eigenvalue of M is

$$\frac{-x - y - z + xyz}{2} \in \{\pm 1\}.$$

The anticommutation $ME = -EM$ then makes the cross term in $(\alpha M + \beta E)^2$ vanish. $\square$

Remark 3.2 (Graph-phase form). If $x = (-1)^a, y = (-1)^b, z = (-1)^c$, then

$$\frac{-x - y - z + xyz}{2} = -(-1)^{ab+ac+bc}.$$

Thus in the joint stabilizer eigenbasis, M is the negative phase associated with the triangle graph, while E flips all three syndrome bits. This explains the four-plus-one support in (2); no graph-state formalism is needed in the proof.

4 The cubic certificate and Yang–Baxter equation

For any $K \in \text{End}(V \otimes V)$, write

$$K_1 = K \otimes \mathbf{1}_V, \quad K_2 = \mathbf{1}_V \otimes K.$$

The Yang–Baxter equation for (3) will follow from a cubic identity for H .

Proposition 4.1. *On the unit circle $\alpha^2 + \beta^2 = 1$,*

$$H(\alpha, \beta)_1 H(\alpha, \beta)_2 H(\alpha, \beta)_1 - H(\alpha, \beta)_2 H(\alpha, \beta)_1 H(\alpha, \beta)_2 = \frac{1}{3} (H(\alpha, \beta)_1 - H(\alpha, \beta)_2)$$

if and only if

$$\beta^2 = \frac{1}{3}, \quad \alpha^2 = \frac{2}{3}.$$

In particular, the four signed points with these squared coordinates are exactly the solutions inside this reflection circle.

Proof. Reduce the left side of (4.1), minus its right side, in the six-letter word basis. Exactly the following 18 words survive:

coefficient	words
$-\alpha(3\beta^2 - 1)/6$	$I I J I J Z, J I Z J I I, Z I J J I I, Z I Z Z I I$
$+\alpha(3\beta^2 - 1)/6$	$I I J I Z J, I I Z I J J, I I Z I Z Z, J I J Z I I$
$-\beta(3\alpha^2 - 3\beta^2 - 1)/3$	$I I X I X X$
$+\beta(3\alpha^2 - 3\beta^2 - 1)/3$	$X I X X I I$
$+\alpha(\alpha^2 - 2\beta^2)/2$	$J I J Z X X, J I Z J X X, X I J X Z J, Z I Z Z X X$
$-\alpha(\alpha^2 - 2\beta^2)/2$	$X I J X J Z, X I Z X J J, X I Z X Z Z, Z I J J X X$

This is a finite certificate: each row follows by repeated use of the displayed 2×2 multiplication table. On $\alpha^2 + \beta^2 = 1$,

$$\alpha^2 - 2\beta^2 = 1 - 3\beta^2, \quad 3\alpha^2 - 3\beta^2 - 1 = 2(1 - 3\beta^2).$$

Hence all coefficients vanish when $3\beta^2 = 1$. Conversely, $\alpha = 0$ or $\beta = 0$ leaves a displayed coefficient nonzero, and when $\alpha\beta \neq 0$ the first and third groups force $3\beta^2 = 1$. The claim follows. $\square$

Choosing

$$\alpha = \sqrt{\frac{2}{3}}, \quad \beta = -\frac{1}{\sqrt{3}}$$

in $H(\alpha, \beta)$ gives exactly (2). Lemma 3.1 and Proposition 4.1 therefore give

$$H^* = H, \quad H^2 = \mathbf{1}_{16}, \quad H_1 H_2 H_1 - H_2 H_1 H_2 = \frac{1}{3} (H_1 - H_2). \quad (5)$$

Every word in (2) contains a traceless factor, so $\text{Tr } H = 0$. The eigenvalues $+1$ and -1 thus both have multiplicity eight.

Proof of the matrix part of Theorem 1.1. Put

$$a = \frac{q-1}{2}, \quad b = \frac{q+1}{2},$$

so $R = a\mathbf{1} + bH$. Since $H_i^2 = \mathbf{1}$, direct expansion gives

$$R_1 R_2 R_1 - R_2 R_1 R_2 = b [a^2(H_1 - H_2) + b^2(H_1 H_2 H_1 - H_2 H_1 H_2)].$$

But

$$\frac{a}{b} = \frac{q-1}{q+1} = \frac{i}{\sqrt{3}},$$

so $a^2 + b^2/3 = 0$. Equation (5) proves the Yang–Baxter equation.

On the $+1$ -eigenspace of H , $R = a + b = q$; on the -1 -eigenspace, $R = a - b = -1$. Therefore

$$(R + \mathbf{1})(R - q\mathbf{1}) = 0, \quad R^* R = \mathbf{1},$$

with both eigenvalues of multiplicity eight. The (-1) -spectral projection is

$$P = \frac{\mathbf{1}_{16} - H}{2},$$

so $P = P^* = P^2$, $\text{rank } P = 8$, and $\eta = 8/16 = 1/2$. Finally, each word in H has a traceless factor on each four-dimensional site. Hence

$$\text{Tr}_1(H) = \text{Tr}_2(H) = 0, \quad \text{Tr}_1(P) = \text{Tr}_2(P) = 2\mathbf{1}_4.$$

□

5 The full localization statement

We now verify that the construction localizes the whole Jones–Wenzl tower, not merely its two-site generator. Let $H_n(q)$ be the specialized Hecke algebra with idempotent generators e_i and braid generators

$$g_i = q\mathbf{1} - (1 + q)e_i.$$

The Yang–Baxter and Hecke relations define a representation

$$\rho_n : H_n(q) \longrightarrow \text{End}(V^{\otimes n}), \quad \rho_n(e_i) = P_i.$$

Let

$$\tau_n(T) = 4^{-n} \text{Tr}(T)$$

be normalized matrix trace.

Proposition 5.1. *The traces $\tau_n \circ \rho_n$ are the Markov traces with parameter $\eta = 1/2$:*

$$\tau_{n+1}((\rho_n(x) \otimes \mathbf{1}_4)P_n) = \frac{1}{2} \tau_n(\rho_n(x)) \quad (x \in H_n(q)).$$

Their kernels equal the trace annihilators. Consequently ρ_n induces an injective representation of the semisimple quotient

$$H_n(3, 6) = H_n(q) / \text{Ann}(\text{tr}_{1/2}).$$

Proof. Taking the partial trace over the last tensor factor and using $\mathrm{Tr}_2(P) = 2\mathbf{1}_4$ gives

$$\begin{aligned}\tau_{n+1}((\rho_n(x) \otimes \mathbf{1}_4)P_n) &= 4^{-(n+1)} \mathrm{Tr}(\rho_n(x) \mathrm{Tr}_{n+1}(P_n)) \\ &= 4^{-(n+1)} 2 \mathrm{Tr}(\rho_n(x)) = \frac{1}{2} \tau_n(\rho_n(x)).\end{aligned}$$

Together with normalization and traciality, this is the defining Markov property. Uniqueness of the Markov trace with prescribed $\mathrm{tr}(e_i) = 1/2$ identifies it with the trace defining $H_n(3, 6)$, as in [2, Theorem 5.28].

Write the trace annihilator as

$$\mathrm{Ann}(\mathrm{tr}_{1/2}) = \{x : \mathrm{tr}_{1/2}(yx) = 0 \text{ for every } y\}.$$

If x lies in it, take $y = x^*$ to obtain

$$0 = \mathrm{tr}_{1/2}(x^*x) = \tau_n(\rho_n(x)^* \rho_n(x)).$$

Normalized matrix trace is faithful, hence $\rho_n(x) = 0$. The reverse inclusion is immediate. Thus the kernel is exactly the annihilator and the induced map on the quotient is injective. $\square$

Galindo, Hong, and Rowell recall the braid-equivariant isomorphism

$$\mathrm{End}_{\mathcal{C}}(X^{\otimes n}) \cong H_n(3, 6)$$

for the two-dimensional simple generator X of $\mathcal{C}(\mathfrak{sl}_3, 6)$ [2, §5.6]. Proposition 5.1 therefore completes the localization assertion in Theorem 1.1.

6 Minimality and amplification

Proof of Corollary 1.2. A one-dimensional local space cannot carry the required two-eigenvalue generator. Galindo, Hong, and Rowell prove that no two-dimensional ordinary unitary localization exists [2, Lemma 5.26].

It remains to exclude local dimension three. After an irrelevant unit scalar normalization, any such generator is a unitary Hecke R -matrix with spectrum $\{-1, q\}$. Lechner's classification [3, Theorem 3.4] leaves at $q = e^{i\pi/3}$ only Markov parameters $1/3, 1/2, 2/3$. The value $1/2$ is impossible in dimension three because it would require a projection of rank $9/2$. At $1/3$ the representation is of Temperley–Lieb type, while at $2/3$ its complementary projection is of Temperley–Lieb type [3, Lemma 3.5 and the following discussion]. Either case kills a nonzero relation already at three strands.

For an explicit obstruction, put

$$c = \frac{q}{(1+q)^2} = \frac{1}{3}, \quad T = P_1 P_2 P_1 - c P_1.$$

The Temperley–Lieb case kills T . In the exceptional $\eta = 1/2$ trace, however, the calculation in [3, Lemma 3.5] gives

$$\mathrm{tr}_{1/2}(T^*T) = \eta(1-c)(\eta-c) = \frac{1}{18} > 0.$$

Thus T is nonzero in $H_3(3, 6)$, so a Temperley–Lieb representation cannot be faithful. For parameter $2/3$, apply the same argument to the complementary projections $Q_i = \mathbf{1} - P_i$, whose exceptional trace is again $1/2$. Theorem 1.1 supplies dimension four. $\square$

There is also an immediate infinite family. If $\mathbf{1}_m$ denotes the identity Yang–Baxter operator on $\mathbb{C}^m \otimes \mathbb{C}^m$, the reshuffled tensor product $R \boxtimes \mathbf{1}_m$ is a unitary Yang–Baxter operator on a base space of dimension $4m$ [3, §3]. Its two eigenvalues and normalized spectral rank are unchanged. Therefore

$$\left[e^{i\pi/3}, \frac{1}{2}, 4m \right] \neq \emptyset \quad (m \geq 1). \quad (6)$$

This does not settle the remaining base dimensions $6, 10, 14, \dots$

7 A (3,2)-generalized normal form

The visible identity in the second qubit of every word in (2) does have a precise chain-level interpretation, but a coordinate swap is essential. Write the i -th four-dimensional site as

$$V_i = \mathbb{C}_{a_i}^2 \otimes \mathbb{C}_{b_i}^2,$$

and swap the two qubits inside every site. In the resulting order

$$(b_1, a_1, b_2, a_2, \dots),$$

the conjugated two-site operator factors as

$$\tilde{R} = \mathbf{1}_{b_i} \otimes K_{a_i, b_{i+1}, a_{i+1}},$$

where

$$K = \frac{q-1}{2} \mathbf{1}_8 + \frac{q+1}{2} K_H$$

and

$$K_H = -\frac{ZZZ + ZJJ + JJZ - JZJ}{\sqrt{6}} - \frac{XXX}{\sqrt{3}}. \quad (7)$$

Adjacent copies of K act on triples shifted by two qubits. Factoring out the global spectator b_1 , the ordinary equation (1) becomes exactly

$$(K \otimes \mathbf{1}_4)(\mathbf{1}_4 \otimes K)(K \otimes \mathbf{1}_4) = (\mathbf{1}_4 \otimes K)(K \otimes \mathbf{1}_4)(\mathbf{1}_4 \otimes K) \quad (8)$$

on $(\mathbb{C}^2)^{\otimes 5}$. Thus K is a (3,2)-generalized Yang–Baxter operator in the convention of [2]. Far commutativity is automatic because nonadjacent triples are disjoint.

More precisely, on n ququarts the sitewise swap gives a unitary conjugacy

$$\tilde{\rho}_R^{(n)} = \mathbf{1}_{b_1} \otimes \rho_K^{(n)}$$

on $\mathbb{C}_{b_1}^2 \otimes (\mathbb{C}^2)^{\otimes (2n-1)}$, where $\rho_K^{(n)}$ is the (3,2)-generalized braid representation. Tensoring by a spectator identity does not change the kernel. Proposition 5.1 therefore also proves that K gives a faithful unitary (3,2)-localization of $H_n(3,6)$.

Equivalently, any (3,2)-generalized operator K on $W^{\otimes 3}$ blocks into an ordinary operator $\mathbf{1}_W \otimes K$ on the local space $W \otimes W$, in the displayed ordering; the ordinary braid relation is $\mathbf{1}_W$ tensored with (8). This elementary blocking observation explains why the present 8×8 active operator yields a 16×16 ordinary one.

The known operator in [2, Eq. (5.2)] is instead of type (3,1) and comes from a quaternionic realization. No equivalence between that generalized localization and (7) is claimed here. The same paper also records a different (3,2) operator in Example 3.14. Its two eigenvalues have projective ratio i (up to inversion), whereas the ratio for K is $-q = e^{-2\pi i/3}$ (up to inversion). The two local operators are therefore not unitarily conjugate up to a phase. In particular, being of type (3,2) is not by itself the new feature here; the exceptional Hecke spectrum and faithful (3,6) localization are.

8 Verification and provenance

The public source package contains three verification routes.

1. `verify_supplied.py` is the original SymPy attachment, preserved byte for byte. It reconstructs the matrices and checks the main identities exactly.
2. `verify_exact.py` is an independent implementation using only the Python standard library. It performs sparse exact arithmetic in $\mathbb{Q}(\sqrt{2}, \sqrt{3}, i)$, directly checks the 16×16 identities, the 64×64 Yang–Baxter equation, both partial traces, and the 32×32 generalized equation (8).
3. `verify_tensor_words.py` never constructs a matrix. It reduces the abstract Pauli-word algebra and checks the complete 18-word certificate in Proposition 4.1.

The latter two programs were written independently of the supplied checker. All comparisons are exact; floating-point tolerances are not used.

The five-word support was originally found by an AI-assisted structured numerical search in a real Pauli–Clifford basis. The discovery code and random seeds were not retained. Accordingly, this note makes no claim that the search was reproducible, exhaustive, or established uniqueness. The mathematical proof does not use the search: after the support was recognized, the coefficients and all identities were derived and verified exactly.

9 Priority audit and scope

The priority audit used the following boundary.

- Galindo, Hong, and Rowell’s 2012 advance publication [2] studied precisely the $\mathcal{C}(\mathfrak{sl}_3, 6)$ representation sequence. It proved nonexistence in ordinary local dimension two and supplied quasi- and $(3, 1)$ -generalized localizations, not an ordinary dimension-four operator.
- Rowell’s quaternionic construction [4] and the generalized-localization literature [5, 1] supply closely related braid representations but no ordinary representative of the displayed exceptional class found in this audit.
- Lechner’s March 2026 preprint explicitly states that $[e^{i\pi/3}, 1/2, 2d]$ was unknown and that dimension two is empty [3, p. 16].
- Focused searches of arXiv, scholarly indexes, general web results, later papers citing or mentioning the preprint, and public GitHub code found no prior formula equivalent to (2) and no published ordinary dimension-four localization.

The defensible conclusion is therefore that the construction *appears new* and answers the open class identified in [3]. This is not a proof of absolute priority: an unindexed, unpublished, paywalled, differently parameterized, or concurrent equivalent may exist. No author or outside researcher was contacted during the audit, in accordance with the independent-research policy.

The broad ordinary-localization question has been open since the 2012 case study; the exact 16×16 formulation was isolated in 2026. The result does not classify the exceptional family in every even dimension, prove that the displayed matrix is unique, or identify an intertwiner with the known quaternionic $(3, 1)$ model.

AI-assistance and verification disclosure

The construction review, proof architecture, exact programs, manuscript, and source audit were developed with heavy assistance from ChatGPT 5.6 Sol under Alec Kriebel’s direction. Alec Kriebel is the named human author and cannot independently certify the mathematics. The work should be treated as a transparent candidate result until qualified experts have checked the proof, the interpretation of the localization definitions, and the priority audit.

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